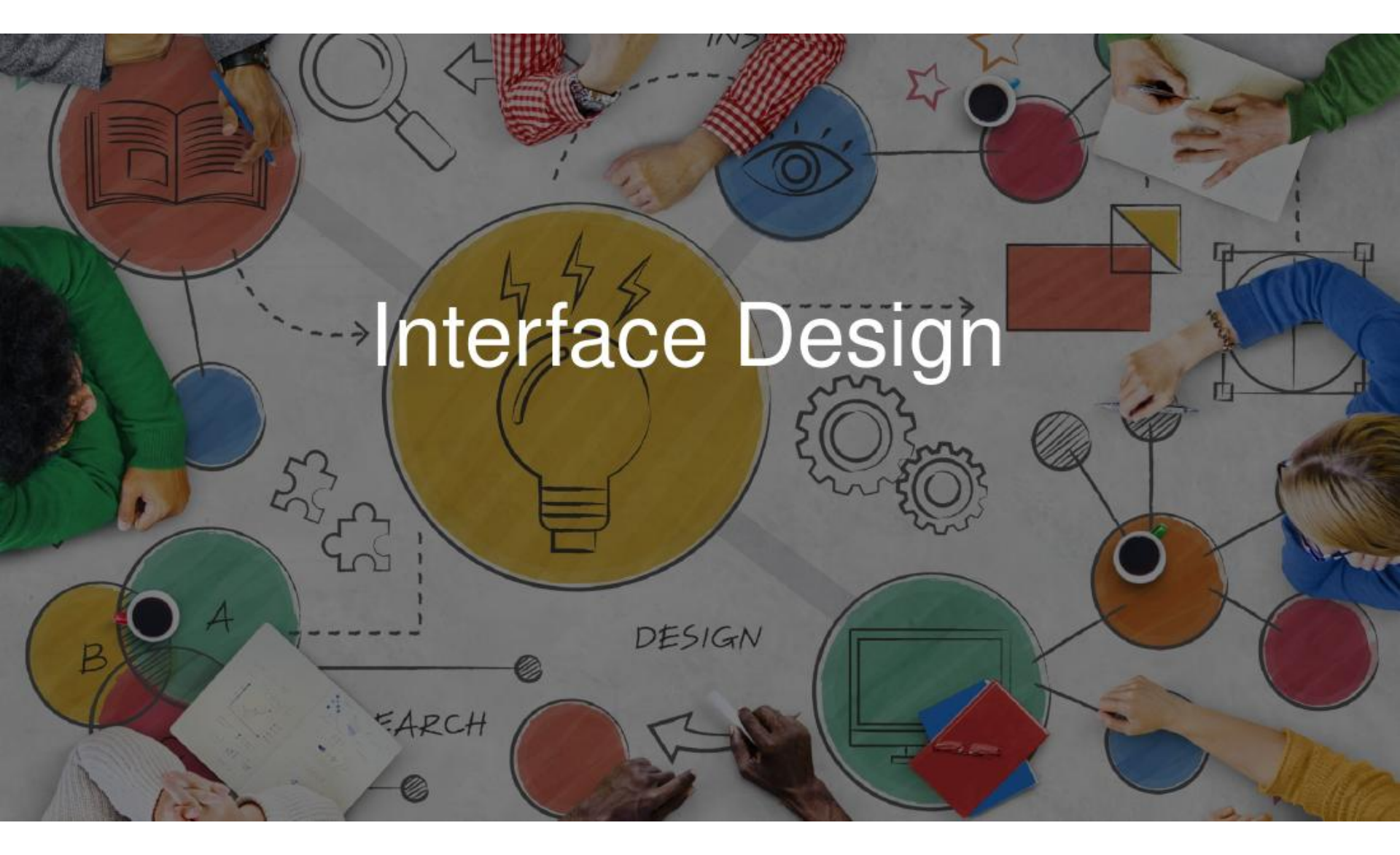


Interface Design

A top-down view of a collaborative workspace. Several people are gathered around a table, their hands and arms visible as they work. The table surface is covered with hand-drawn diagrams and sketches. A large yellow circle in the center contains a lightbulb with three lightning bolts. To the left, a red circle contains an open book. To the right, a blue circle contains an eye. Other elements include a magnifying glass, a star, a gear, a puzzle piece, a computer monitor, and various colored circles and lines. The word "DESIGN" is written in the lower center, and "RESEARCH" is written in the lower left. The overall scene suggests a creative and collaborative design process.

Objectives

- Discuss the importance of prototyping.
- Identify the types of Prototyping.

Introduction

- The interface design is the design of the **user interface** of the system being developed.
- It must be as **efficient and straightforward** as possible.
- The **designers** must have a good understanding of the **user needs and their interaction** with other components.
- The activities include **prototyping** of the user interface and **usability testing** of the prototypes.

What do you think is Prototyping?



PROTOTYPING

PHASE



Introduction

- ***Prototyping*** is an activity that involves creating a scaled-down or incomplete version of the system being developed **to demonstrate or test aspects** of it.
- The reasons to do prototyping include **aid** in the design, **basis** for coding & testing, and **allows interaction with users to ensure satisfaction**.



A **prototype** is an early sample, model, or release of a product built to **test a concept or process.**

8 Golden Rules When Designing User Interface

According to **Sneiderman**

1



Strive for
consistency



2



Give shortcuts
to the user



3



Offer informative
feedback



4



Make each interaction
with the user yield
a result



5



Offer simple
error handling



6



Permit easy
undo of actions



7



Let the user
be in control



8



Reduce short-term
memory load on
the user





10 common UI Design Mistakes

- 10 common UI design mistakes according to careerfoundry.com

Link:

- <https://careerfoundry.com/en/blog/ui-design/common-ui-design-mistakes/>

Two (2) kinds of Prototypes

- There are two kinds of prototypes that will be presented in this lesson: low-fidelity (Lo-Fi) and high fidelity (Hi-Fi) prototypes.

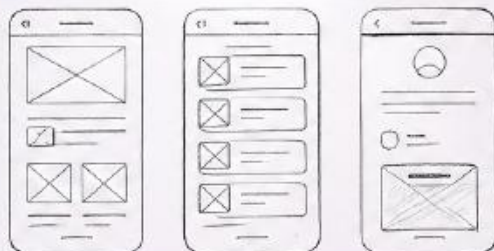
What do you think is **PROTOTYPING**?



Prototyping is the process of creating a sample or early version of a product to test ideas, design, and functionality before making the final version.

★ It helps you see how something works, find problems, and improve it.

LO-FI PROTOTYPE (Low-Fidelity)



- Simple and rough (sketches, paper drawings)
- Focuses on layout and basic ideas
- Fast and cheap to create
- No detailed design or interaction



Example: Drawing an app screen on paper

HI-FI PROTOTYPE (High-Fidelity)



- Detailed and realistic
- Looks like the final product
- Includes colors, fonts, and interactions
- Used for testing user experience



Example: A clickable app design made in Figma

WHY PROTOTYPING IS IMPORTANT?



Helps visualize ideas



Detects errors early



Saves time and cost



Improves design quality

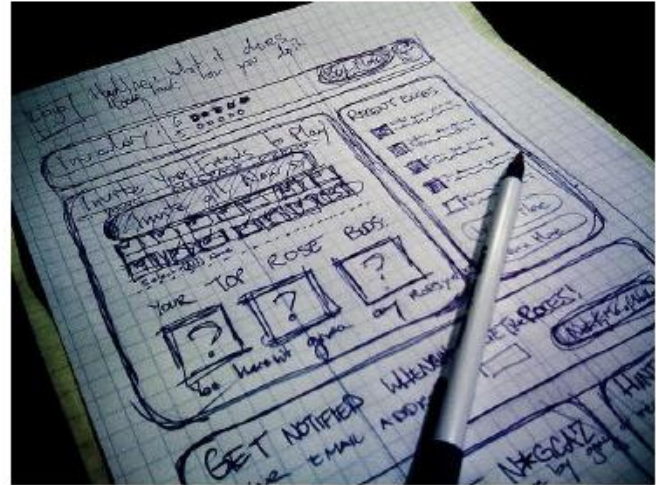
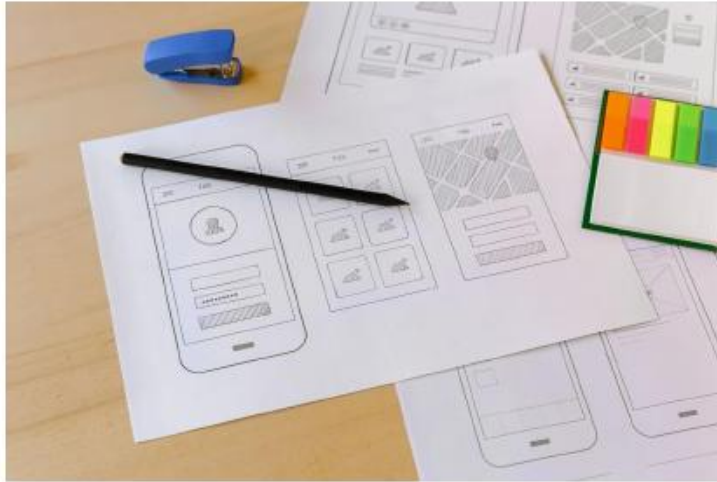
Two (2) kinds of Prototypes

Fidelity explains the level of detail and functionality that the design has. This measure captures the **level of detail** in content, visuals, and interactivity.

Low fidelity pulls in key placeholders, while high fidelity mirrors the final design closely.

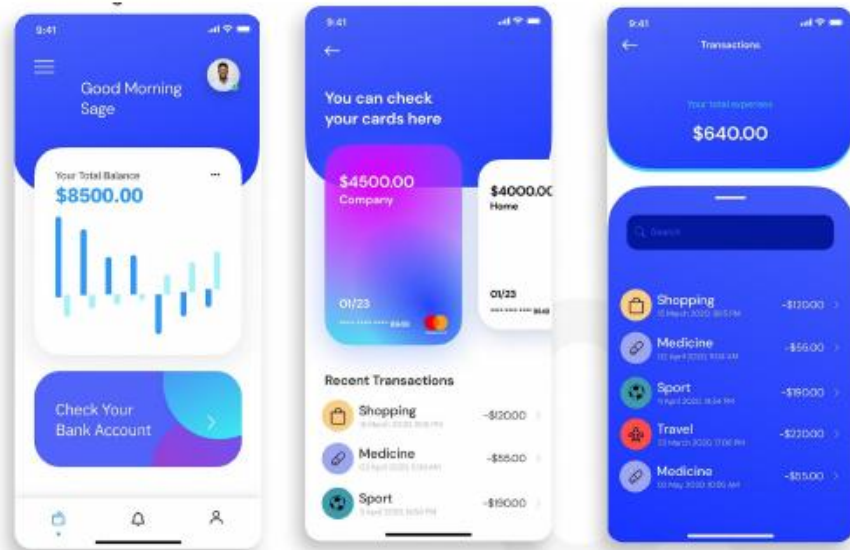
Low fidelity design typically has the basic elements of the final design, while high fidelity looks as close as possible to the final design.

Two (2) kinds of Prototypes



Lo-Fi Prototype

Two (2) kinds of Prototypes

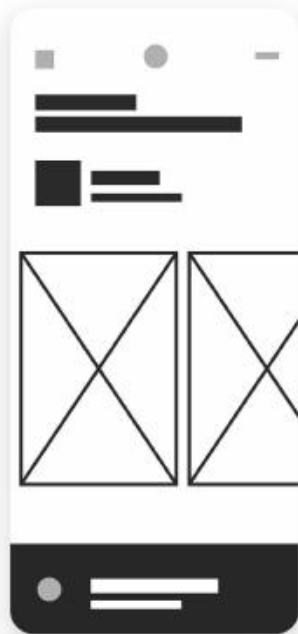


Hi-Fi Prototype

Two (2) kinds of Prototypes

Low Fidelity

High Fidelity



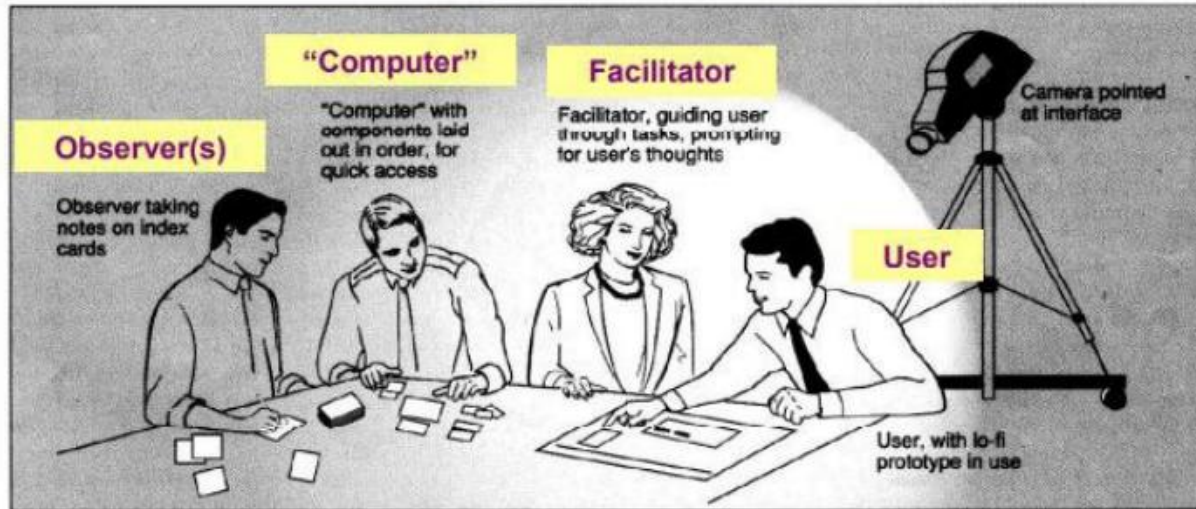
LOW - FIDELITY



HIGH-FIDELITY

Usability Testing

- Usability measures the effectiveness of the user interface with which the users can achieve tasks in one software environment. The paper prototyping usability session is shown below.



What is Usability Testing?

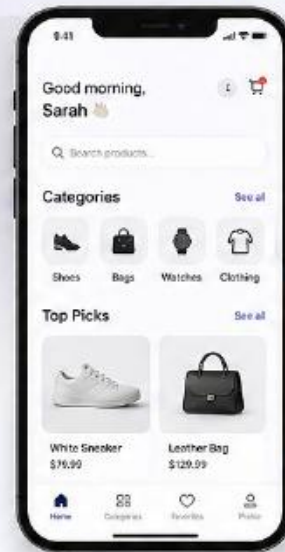


Usability testing is the process of evaluating a product by testing it with **real users**. It helps identify problems and **improve the user experience**.

How it works



Example



Can you find the product you would buy?

Feedback

- Hard to find search
- Add filters on category

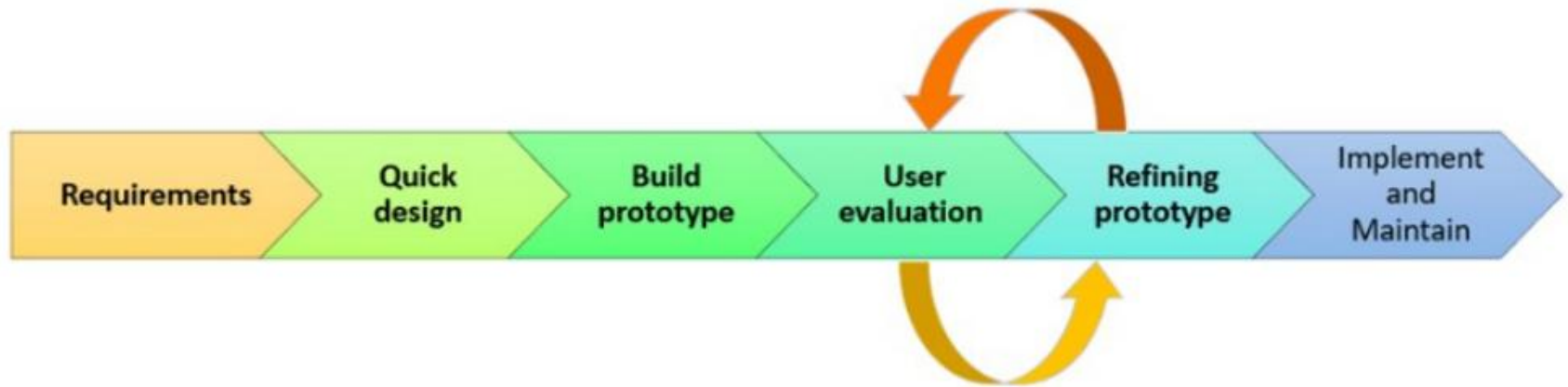
Usability Testing

- There are usability standards that provide guidance on the usability of systems and services.
- The ISO 9241-11:2018(en) Ergonomics of human-system interaction measures the effectiveness, efficiency, and satisfaction of the user experience.

What is Prototyping Model?

- **Prototyping Model** is a software development model in which prototype is built, tested, and reworked until an acceptable prototype is achieved.
- It also creates base to produce the final system or software. It works best in scenarios where the project's requirements are not known in detail.
- It is an iterative, trial and error method which takes place between developer and client.

Phases of Prototyping Model



How do you know your users? What are the steps to understand your users?



Prototyping Model has following six SDLC phases

(SOFTWARE DEVELOPMENT LIFE CYCLE)

STEP 1

Requirements Gathering & Analysis

Understand what the system should do.



Analyze
Requirements



Interview
Users



Outcome



Document
Requirements



Understand
User Needs



Define Goals &
Expectations



Basis for
Prototype Design

STEP 2

Quick Design

Create a simple design to give a basic idea of the system.



Preliminary
Design



Brief Idea
of the System



Helps in
Prototyping



Outcome



Quick
Design



Visualize
the System



Get Early
Feedback



Move to
Prototyping

STEP 3

Build a Prototype

Build a small working model of the system.



Outcome



Design
Prototype



Create Working
Model



Test Basic
Functionality



Prototype
Ready

STEP 4

Initial User Evaluation

Present the prototype to users and gather feedback.



Outcome



Show to
Users



Collect Feedback
& Suggestions



Identify Strengths
& Weaknesses

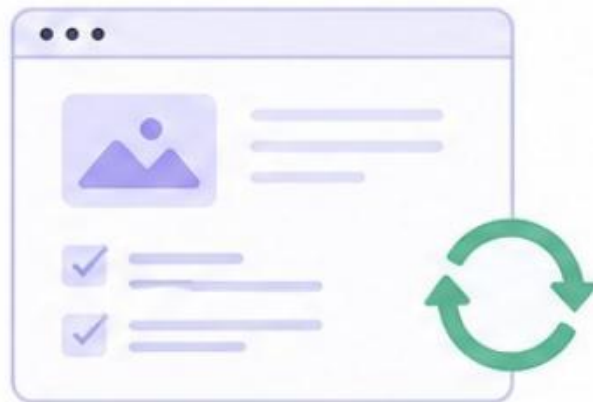


Feedback to
Developers

STEP 5

Refining Prototype

Refine the prototype based on feedback until users are satisfied.



Outcome



Review Feedback



Refine & Improve



User Satisfied



Final Prototype
Approved

STEP 6

Implement Product and Maintain

Develop, deploy, and maintain the final system.



Outcome



Develop Final System



Test & Deploy to Production



Maintain System



Ensure Stability & Performance

Prototyping Model



The Process of Interaction Design



Overview

- What is Interaction Design?
 - Four basic activities
 - Three key characteristics
- Some practical issues
 - Who are the users?
 - What are 'needs'?
 - Where do alternatives come from?
 - How do you choose among alternatives?
- Lifecycle models from software engineering
- Lifecycle models from HCI



What is Interaction Design?

What is Interaction Design?

- It is a process:
 - a goal-directed problem-solving activity informed by intended use, target domain, materials, cost, and feasibility
 - a creative activity
 - a decision-making activity to balance trade-offs

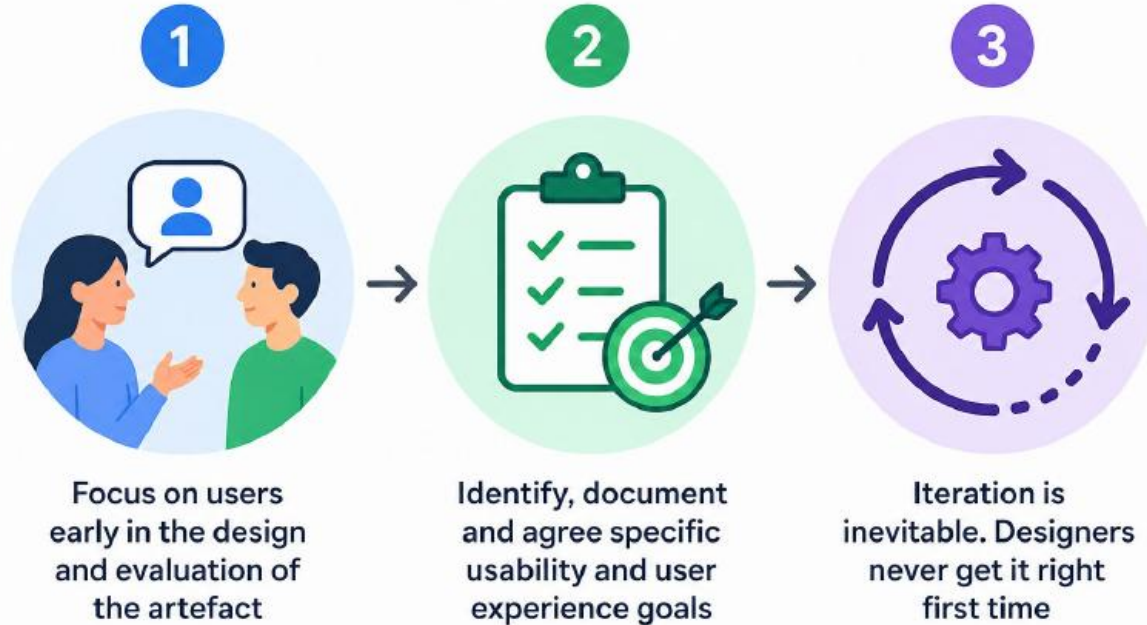
What is Interaction Design?

- It is a representation:
 - a plan for development
 - a set of alternatives and successive elaborations

Four basic activities

1. Identifying needs and establishing requirements
2. Developing alternative designs
3. Building interactive versions of the designs
4. Evaluating designs

Three key characteristics



Some practical issues

- Who are the users?
- What are needs?
- Where do alternatives come from?
- How do you choose among alternatives?

Who are the users/stakeholders?



Those who
interact directly
with the product



Those who
manage
direct users



Those who
receive output
from the product



Those who
make the
purchasing
decision



Those who
use competitor's
products

Three categories of user (Eason, 1987)

1 Primary



Frequent
hands-on

2 Secondary



Occasional
or via
someone else

3 Tertiary



Affected by its
introduction, or
will influence its
purchase

Who are the stakeholders?



What are the users' capabilities?

Humans vary in many dimensions:

- size of hands may affect the size and positioning of input buttons
- motor abilities may affect the suitability of certain input and output devices
- height if designing a physical kiosk
- strength - a child's toy requires little strength to operate, but greater strength to change batteries



What are needs?



- Users rarely know what is possible



- Users can't tell you what they 'need' to help them achieve their goals



- Instead, look at existing tasks:
 - 📍 their context
 - 📘 what information do they require?
 - 👥 who collaborates to achieve the task?
 - ❓ why is the task achieved the way it is?



- Envisioned tasks:
 - 🔄 can be rooted in existing behaviour
 - 💡 can be described as future scenarios

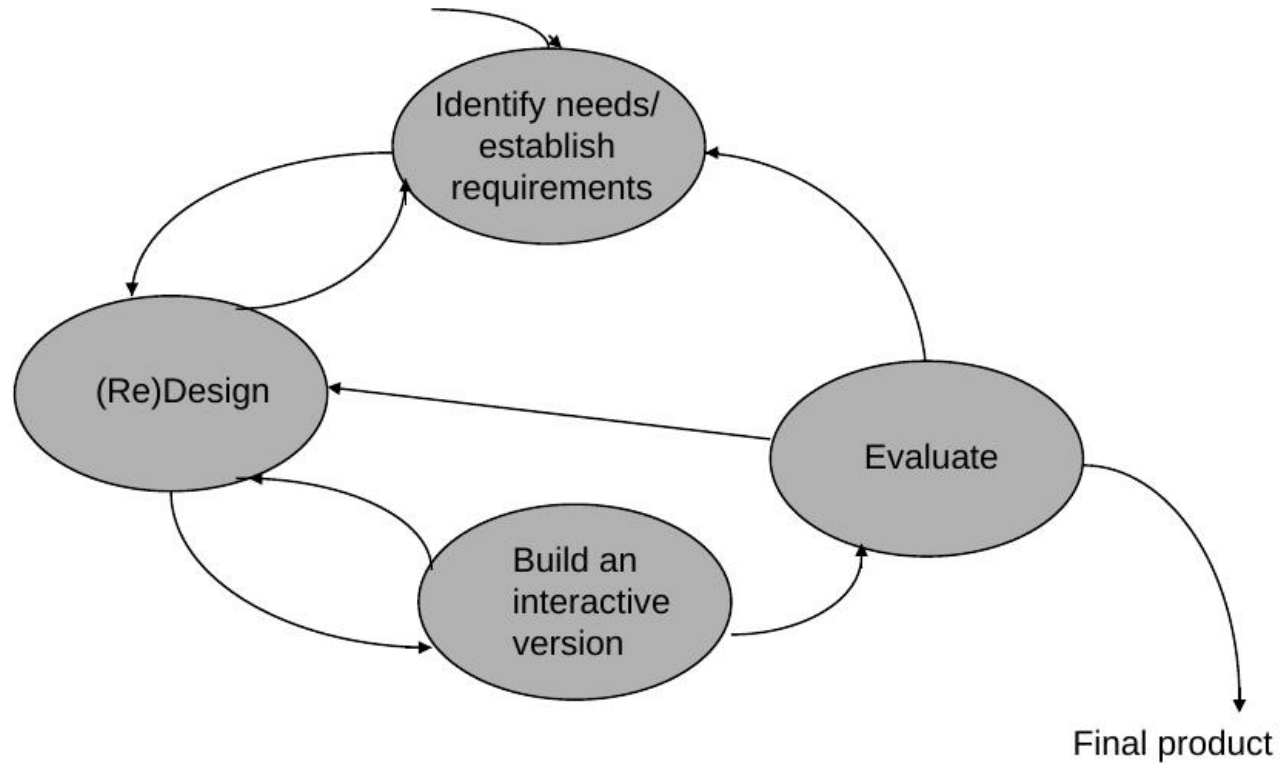


Lifecycle Models

Lifecycle models

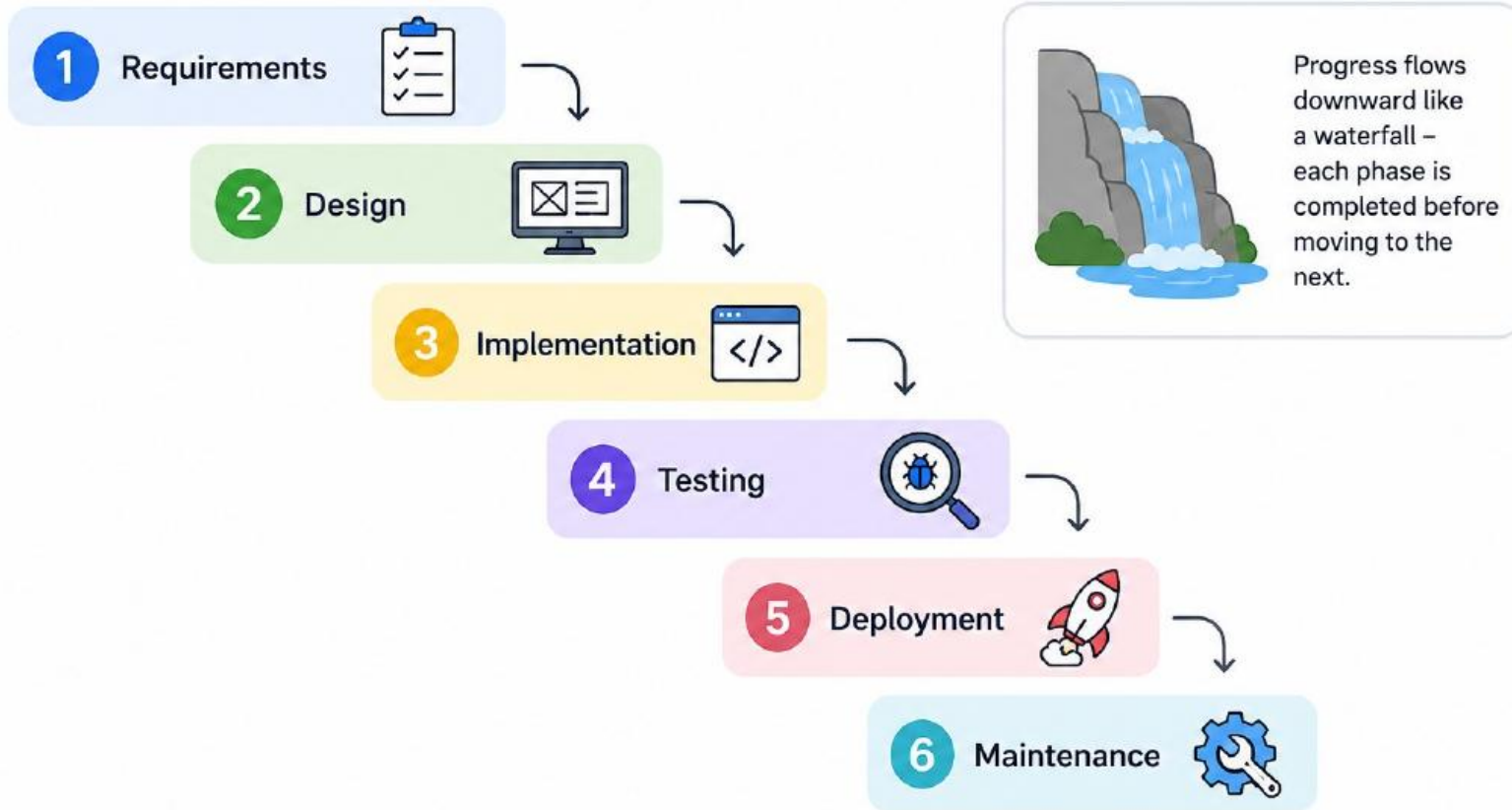
- Show how activities are related to each other
- Lifecycle models are:
 - management tools
 - simplified versions of reality
- Many lifecycle models exist, for example:
 - from software engineering: waterfall, spiral, JAD/RAD, Microsoft
 - from HCI: Star, usability engineering

A SIMPLE INTERACTION DESIGN MODEL



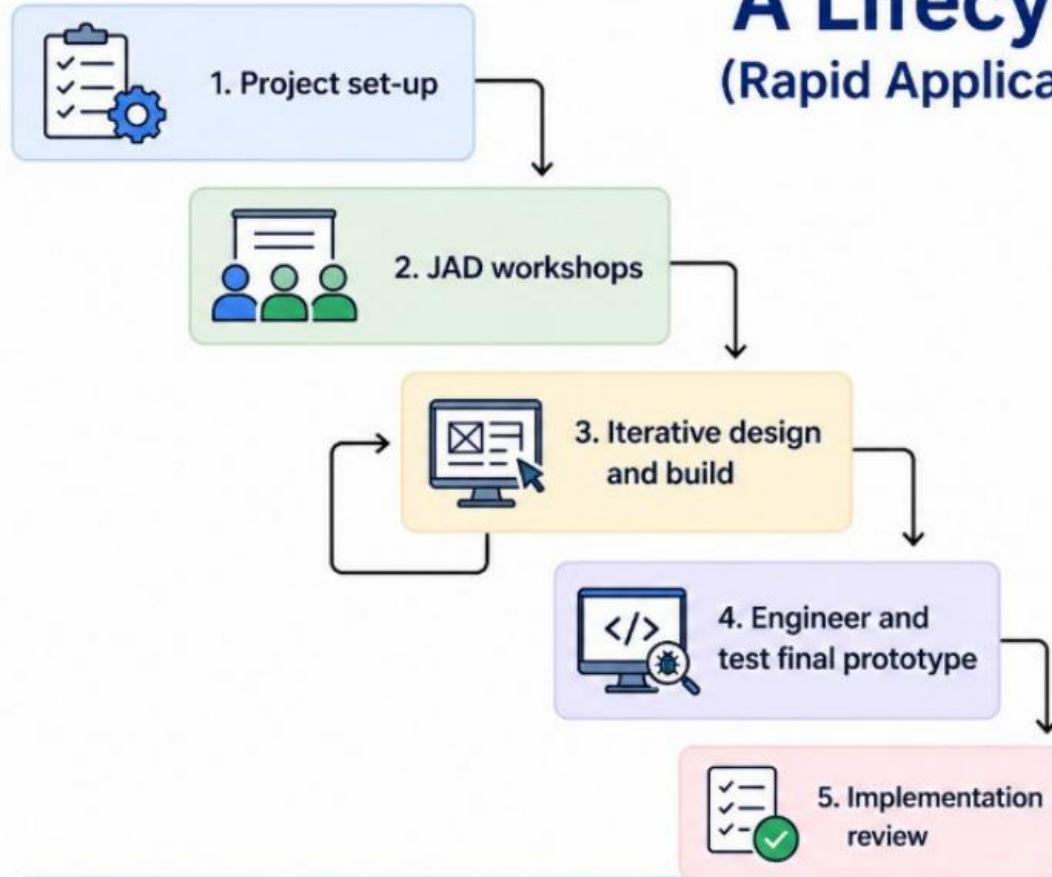
Exemplifies a user-centered design approach

Waterfall Model



A Lifecycle for RAD

(Rapid Applications Development)



When to use RAD



When rapid delivery is important



When users are available and can actively participate



When requirements are clear and at a high level



For projects with a short time frame



For systems that can be modular and iterative

Spiral Model (Barry Boehm)

Important features



Risk analysis

Identify and assess risks early and continuously.



Prototyping

Build prototypes to explore and validate ideas.



Iterative framework

Ideas are checked and evaluated in each cycle.

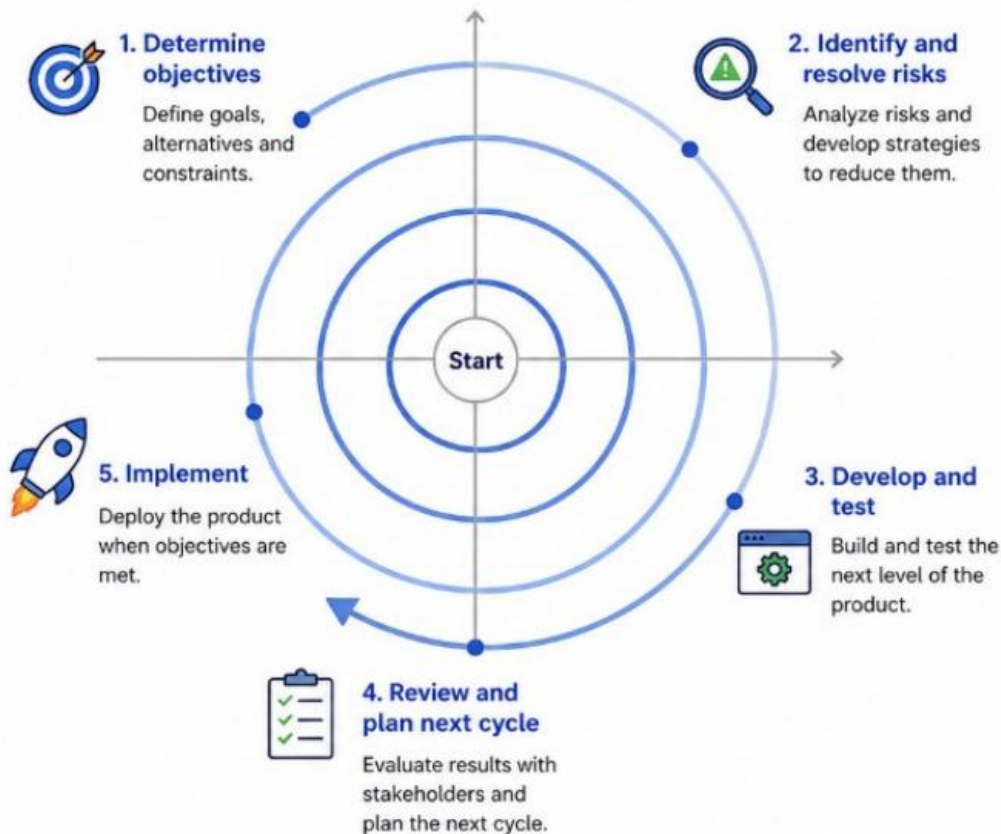


Encourages alternatives

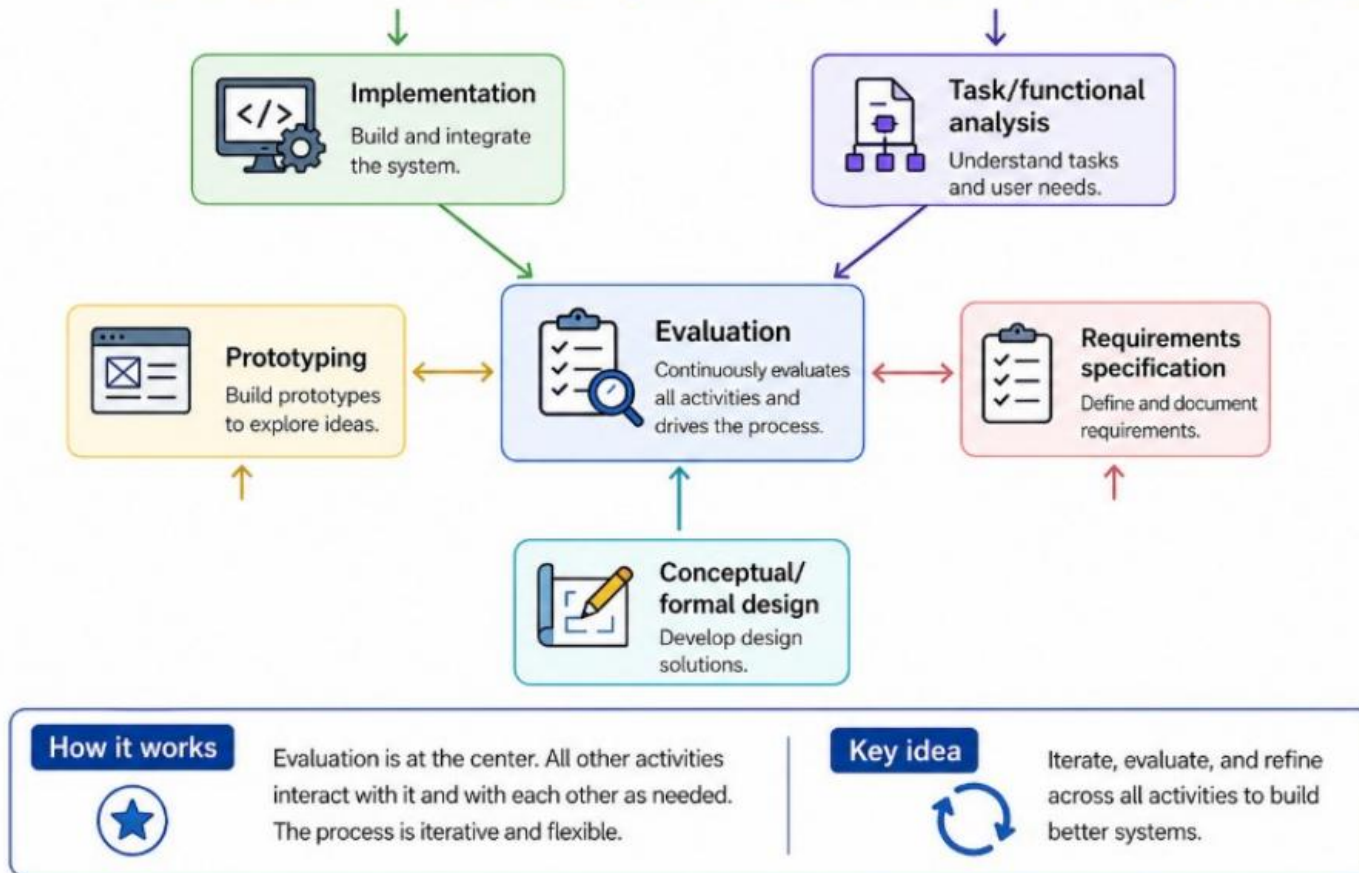
Multiple solutions and approaches are considered.



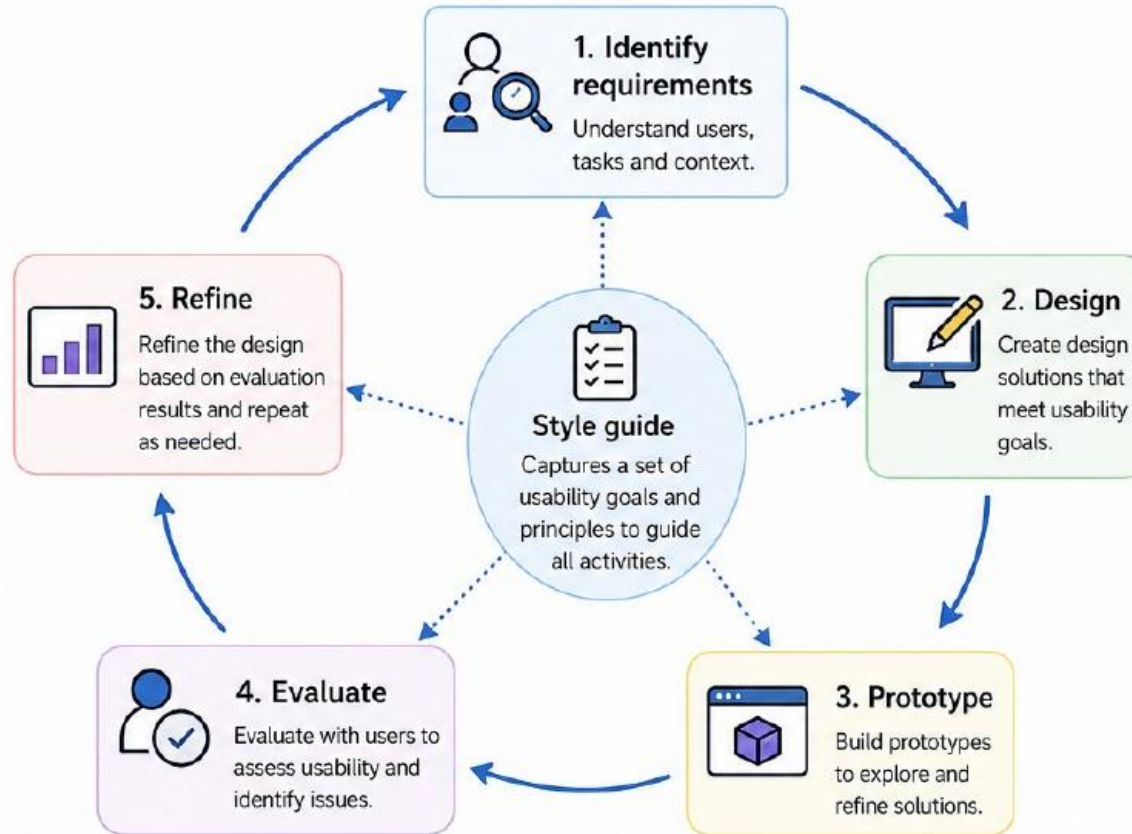
Good for large and complex projects but not simple ones.



The Star Model (Hartson and Hix, 1989)



Usability engineering lifecycle model



Summary

Four basic activities in the design process

1. Identify needs and establish requirements
2. Design potential solutions ((re)-design)
3. Choose between alternatives (evaluate)
4. Build the artefact

These are permeated with three principles

1. Involve users early in the design and evaluation of the artefact
2. Define quantifiable & measurable usability criteria
3. Iteration is inevitable

Lifecycle models show how these are related